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Review

Assertive communication training for nurses to speak up in cases of medical errors: A systematic review and meta-analysis

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ABSTRACT

Background: Several authors have previously assessed the effects of assertive communication training for nurses to speak up in cases of medical errors. Inconsistent results regarding the nurses' attitudes, behaviors, and confidence levels were noticed.
Objective: To identify the effectiveness of assertive communication training on nurses' behaviors, attitudes, and confidence levels for speaking up in cases of medical errors and to identify vital components for success.
Design: Systematic review and meta-analysis.
Data sources: PubMed, ERIC, Embase, Scopus, and CINAHL were searched up for studies published from the inception of the database to December 16, 2022.
Methods: Two researchers independently performed a primary screening of titles and abstracts for relevant studies, followed by a review of full texts if the references met inclusion criteria and quality assessment. Data were retrieved for nurses and nursing students who received medical error–related assertive communication training for speaking up, and learning outcomes for attitudes, behaviors, and confidence levels were reported based on pooled data. Pooled estimates were calculated using a random-effects model.
Results: A total of eleven studies with 1299 participants were included in systematic review, and among them nine studies with 804 participants were analyzed in meta-analysis. From the pooled results, the intervention group, which received assertive communication training, exhibited substantially improved speaking-up behaviors compared with the control group (SMD = 0.58; 95 % CI, 0.14–1.03). Considerable differences were noted in the nurses' times of speaking up in cases of medical errors between pretest (38 %; 95 % CI, 0.14–0.68) and posttest (78 %; 95 %CI, 0.70–0.85) based on the pooled data. The nurses' attitudes and confidence levels for speaking up varied markedly between pretest and posttest.
Conclusion: Structured assertive communication training may improve nurses' speaking-up behaviors in cases of medical errors. To conduct effective assertive communication training, nursing educators should incorporate multiple teaching approaches into structured training and ensure an adequate training duration.

1. Introduction

A total of 251,000 medical error–related deaths are reported annually in the United States, thus making medical errors the third leading cause of death in this country (Makary and Daniel, 2016). People may die or be permanently injured because of these errors. Many medical error–related adverse events are preventable if identified and stopped early (Austin and Derk, 2016). *Speaking up* in cases of medical errors can be described as assertive communication to defend or protest patient safety concerns immediately by asking questions, delivering information, or exchanging opinions until a clear resolution is obtained (Schwappach and Richard, 2018; Schwappach and Gehring, 2014).

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Healthcare providers can effectively reduce harm to patients by speaking up (Makary and Daniel, 2016).

According to Reason's (2005) "Swiss cheese model", there are multiple layers of defense to prevent errors from reaching patients. In the acute environment, nurses are the first-line direct care providers, and they can play a significant role in stopping patient harm if they speak up in the situation of medical errors. The speaking-up behavior of nurses is positively correlated with reducing potential medical errors and enhanced patient safety (Sayre et al., 2012a, 2012b). In one study, although 84 % of the total nurses noticed potentially unsafe practices, only 31 % spoke up about their concerns (Maxfield et al., 2011). This highlights the importance of interventions aimed at helping nurses overcome the barriers to speaking up. Some of these barriers include low confidence (Raica, 2009), hierarchy among healthcare professionals (Green et al., 2017), the clinical context, perceived safety after speaking up, and perceived efficacy of speaking up (Okuyama et al., 2014). A positive attitude and competence are essential for nurses to speak up when they have patient safety concerns for preventing harm to patients.

In the healthcare system, assertive communication facilitates speaking up in cases of patient safety concerns (AHRQ, 2019). Nurses may intercept potential medical errors and enhance patient safety through assertive communication (Sayre et al., 2012a, 2012b). Assertive communication training, an educational intervention, can help address the patient safety concerns of healthcare providers by improving their behaviors, attitudes, and confidence levels for speaking up in cases of medical errors (Best and Kim, 2019; Kent et al., 2015; Omura et al., 2019a, 2019b; Oner et al., 2018; Sayre et al., 2012a, 2012b; Shanks et al., 2020). The Kirkpatrick Model is a globally recognized method for evaluating the outcomes of educational programs (Level 1, reactions; Level 2, learning; Level 3, behavior changes; and Level 4, results). This model (Kirkpatrick, 2009) can be used to evaluate post-training improvements in trainees and the level of educational outcomes.

There are inconsistent results in the literature regarding the attitudes, behaviors, and confidence levels of nurses after interventions for speaking up; these parameters may be influenced by various factors, such as educational content, teaching approaches, intervention duration, and sample size. We conducted this review to better understand the effectiveness and essential components of assertive communication training. The aim of this systematic review and meta-analysis was to assess, appraise, and synthesize the effectiveness of the assertive communication training in improving the behaviors, attitudes, and confidence levels of nurses and nursing students for speaking up in cases of medical errors.

2. Methods

2.1. Protocol and registration

The review protocol is registered in the International Prospective Register of Systematic Reviews (No. CRD42021242175) and reporting is in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines (Moher et al., 2009).

2.2. Search strategy

We searched selected keywords on databases, such as PubMed, Education Resources Information Center, Embase, Scopus, and CINAHL, for relevant studies from the inception of the database to December 16, 2022. The search was conducted without any language restriction. If the search results included studies not written in English, the research team used translation software and then evaluated the studies' eligibility based on the inclusion and exclusion criteria. The search strategy was as follows: (teach* OR taught OR learn* OR course* OR educat* OR instruct* OR lesson OR program OR class OR train* OR coach*) AND (speak up OR speaking up OR assert* communicat* OR silence) AND (medical errors OR error OR medical mistakes OR adverse event OR harm* OR patient harm OR medical incidents OR medical malpractice) AND (nurses OR nursing student OR physician OR doctors OR medical student). In addition, the end-list references of the identified studies and systematic reviews were assessed to further identify potentially eligible studies (Supplementary 1) (Higgins et al., 2022).

2.3. Eligibility criteria

The inclusion criteria were set based on the PICO (population, intervention, comparison, and outcome), defined as follows (Higgins et al., 2022):

- Population: registered professional nurses, nursing students, and nurse members of multidisciplinary teams
- Intervention: assertive communication training for speaking up in cases of medical errors (no restriction in terms of pedagogy and study design)
- Comparison: nurses not receiving this training
- Outcome: learning outcomes of the speaking-up training (knowledge, attitudes, behaviors, and confidence levels)

Studies were excluded if 1. there was no specific nurse learning outcome reported in multidisciplinary team study; 2. the content of the intervention was not related to medical errors or patient safety concerns; 3. the full text was not available; 4. the searched studies were protocols, reviews, or conference abstracts; 5. the data was insufficient for conducting analysis such as only p-value was provided in some articles.

2.4. Study selection and data extraction

Based on the searching strategy, two researchers (HWC and YRC) independently screened titles and abstracts from five databases for relevant studies and reviewed full texts if the references met inclusion criteria. Using a predesigned extraction form, key information and outcome data from the eligible studies were extracted independently by each reviewer. Data extracted included first author's surname, publication year, study country, study design, sample size, study participants, and intervention program (information on teaching approaches, training content, intervention length, measurement tools, and outcome assessment time). Data on the outcomes of interests were carefully extracted and checked twice. Speaking-up-related parameters were assessed by self-reports and faculty observations. Data were handled as continuous variables, such as mean and standard deviation (SD), and categorical variables such as the number of times nurses spoke up about medical errors in each group within the included studies. We reported the number of trainees in each group within the included studies. Any disagreement regarding study selection and data extraction between the two reviewers was resolved through discussion with a third reviewer (YNK).

2.5. Quality assessment

The two reviewers (HWC and YRC) independently assessed the quality of each eligible study using the Joanna Briggs Institute (JBI) critical appraisal tool. All questions were rated as "yes", "no", or "not applicable". Because the total number of studies was limited, we did not exclude studies from our review even when appraisal questions were answered with "no" or "not applicable". All disagreements were resolved through discussion and consultation with a third reviewer (YNK) for consensus.

2.6. Statistical analysis

Data on speaking-up scores and confidence levels were pooled in random-effects models using R and Studio version 1.0.153 with the metafor package (Viechtbauer, 2010). Because of the large amount of

data from single-arm studies, self-reported speaking-up scores and confidence levels were pooled using the *metamean* function with study name, trainee numbers, and mean and SD values. For data on observed speaking-up behaviors from single-arm studies, the *metaprop* function was used. Data obtained from the intervention groups in two-arm studies were also included in single-arm data synthesis. In addition to a single-arm meta-analysis, to ensure successful data synthesis for speaking-up behaviors in two-arm comparisons, the standardized mean difference (SMD) of the data reported by Omura (Omura et al., 2019a) was derived using the log odds ratio according to the formula stated in the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins et al., 2022). Point estimates with 95 % confidence intervals (CIs) were obtained. In addition, in data synthesis, I^2 was calculated to determine heterogeneity across the studies.

3. Results

3.1. Study selection

The initial literature search returned a total of 4491 articles. In total, 1126 studies were excluded because of duplication, and 3365 studies were excluded based on the exclusion criteria. Although full texts could be retrieved from a total of 20 eligibility studies, 9 of them were excluded because the intervention was not related to speaking up, no outcomes of nursing professionals were reported, or the content was not related to medical error or patient safety.

The remaining 11 studies were included in the systematic review. Among these, 9 studies were included in the meta-analysis (Fig. 1) (Da Silva et al., 2020; Doherty et al., 2016; Kent et al., 2015; Omura et al., 2019a, 2019b; Oner et al., 2018; Sawyer et al., 2013; Sayre et al., 2012a;

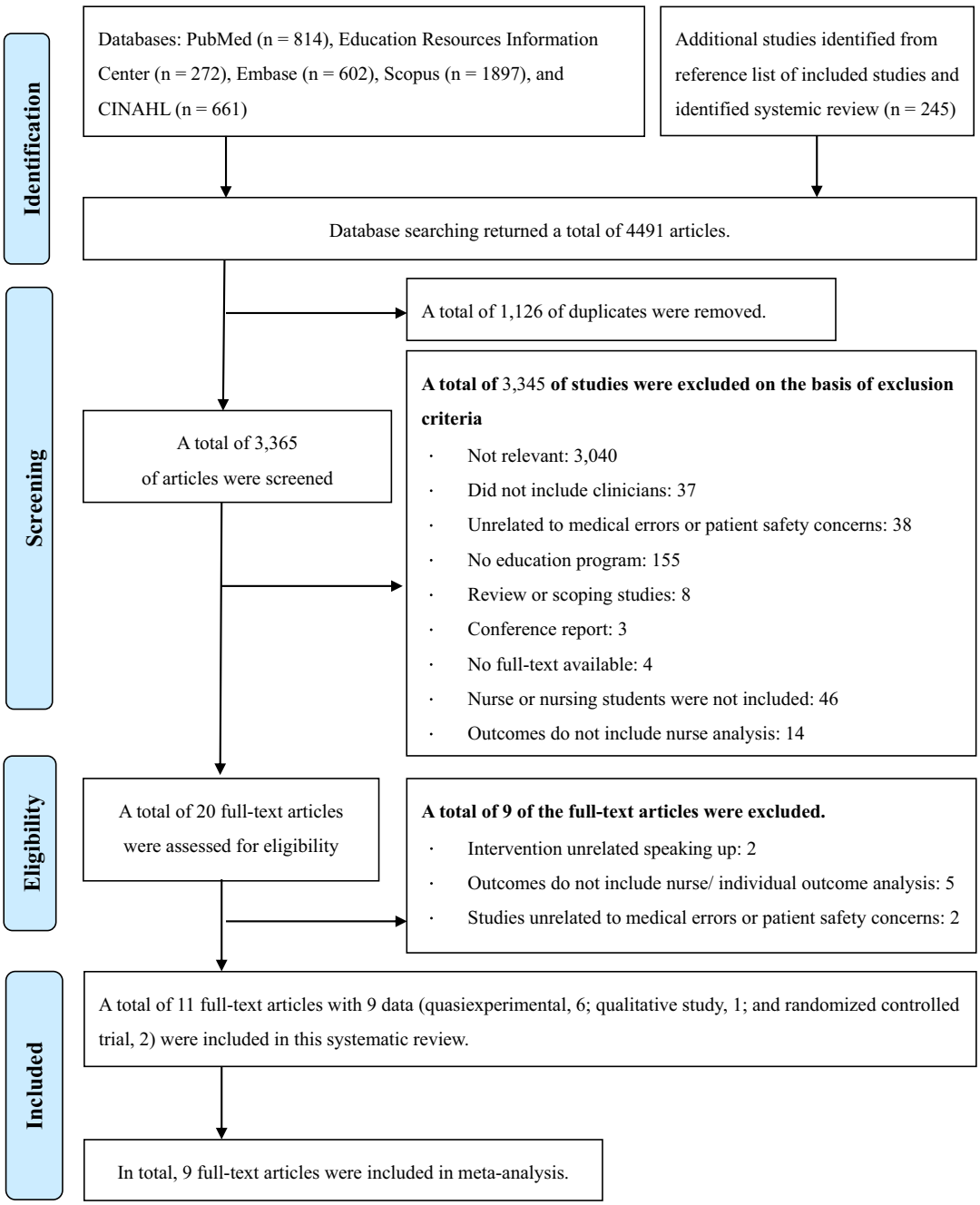

Fig. 1. Flowchart for study selection.

Shanks et al., 2020). Two studies were excluded from the meta-analysis because one of them was a qualitative study (Hanson et al., 2020) and another one only contained a p-value without suitable extractable data (Sayre et al., 2012b).

3.2. Study characteristics

The 11 studies selected for this review were published between 2012 and 2020. Eight studies were conducted in the United States, and the others were in Canada, Japan and Australia, respectively. For the study design, eight studies were quasi-experimental design, two were RCTs (Da Silva et al., 2020; Oner et al., 2018), and one was a qualitative study (Hanson et al., 2020). For types of participants, registered nurses participated in four studies (Oner et al., 2018; Sawyer et al., 2013; Sayre et al., 2012a, 2012b), nursing students participated in seven studies (Da Silva et al., 2020; Doherty et al., 2016; Hanson et al., 2020; Kent et al., 2015; Omura et al., 2019a, 2019b; Shanks et al., 2020), and a multidisciplinary team was enrolled in one study (Sawyer et al., 2013). The registered nurses were aged 29–53 years (Oner et al., 2018; Sawyer et al., 2013), whereas the nursing students were aged 18–34 years (Da Silva et al., 2020; Doherty et al., 2016; Kent et al., 2015; Omura et al., 2019a, 2019b). The sample sizes ranged from 42 to 535, with a total number of 1299 participants (Da Silva et al., 2020; Doherty et al., 2016; Hanson et al., 2020; Kent et al., 2015; Omura et al., 2019a, 2019b; Oner et al., 2018; Sawyer et al., 2013; Sayre et al., 2012a, 2012b; Shanks et al., 2020). The teaching approaches included simulation (Shanks et al., 2020), virtual simulations (Da Silva et al., 2020), discussion (Sayre et al., 2012a, 2012b), multi-method with simulation (Oner et al., 2018; Sawyer et al., 2013), and multi-method without simulation (Doherty et al., 2016; Hanson et al., 2020; Kent et al., 2015; Omura et al., 2019a, 2019b). The lengths of interventions varied from 1.5 h (Omura et al., 2019a, 2019b) to 6 h (Oner et al., 2018; Sawyer et al., 2013; Sayre et al., 2012a, 2012b). Authors in five studies did not provide information on the overall length of the intervention (Da Silva et al., 2020; Doherty et al., 2016; Hanson et al., 2020; Kent et al., 2015; Shanks et al., 2020). The outcome checkpoint times were pre- and posttests in seven studies (Doherty et al., 2016; Kent et al., 2015; Oner et al., 2018; Sawyer et al., 2013; Sayre et al., 2012a, 2012b; Shanks et al., 2020) and posttest in four studies (Da Silva et al., 2020; Hanson et al., 2020; Omura et al., 2019a, 2019b). Self-reported questionnaires were identified in eight studies (Doherty et al., 2016; Hanson et al., 2020; Kent et al., 2015; Omura et al., 2019a, 2019b; Sayre et al., 2012a, 2012b; Shanks et al., 2020) and the use of objective observations of speaking-up behavior was identified in two studies (Da Silva et al., 2020; Sawyer et al., 2013). Dual assessment of both self-reported willingness to speak up and observed speaking-up behavior was reported in only one study (Oner et al., 2018). Table 1 summarizes the characteristics of the included studies.

3.3. Risk of bias

The risks of bias were assessed using the JBI critical appraisal checklists. For two RCTs, we focused on the potential bias in RCT processes. The participants were stratified by their work departments and were then assigned to groups through block randomization. Group assignment was performed using computer-generated sequentially numbered allocation cards in opaque sealed envelopes (Oner et al., 2018), and each group through a simple random allocation using software (Da Silva et al., 2020). Both study outcome assessors were blinded to group allocation. Some items did not satisfy all criteria of the JBI critical appraisal tool; for example, the participants and instructors were not blinded to training programs (Oner et al., 2018), and the results of the intention-to-treat analysis was not reported because some participants were unavailable owing to different work shifts or refused to participate after randomization (Oner et al., 2018) and increased academic workload or personal issues (Da Silva et al., 2020).

For the eight quasi-experimental studies, we focused on the possibility of bias in its design, conduct and analysis. Designs and statistical approaches that addressed the possible bias at baseline were used in three of the included studies (Shanks et al., 2020; Sayre et al., 2012a, 2012b). The participants were senior nursing students from the same university (Shanks et al., 2020), and regression analysis was performed to control for group mean differences (Sayre et al., 2012a, 2012b). Most quasi-experimental studies did not adequately report their strategies to account for the loss of the participants to follow-up (Doherty et al., 2016; Omura et al., 2019a, 2019b; Sayre et al., 2012a, 2012b; Shanks et al., 2020).

For the qualitative study (Hanson et al., 2020), there was no sufficient information to assess the philosophical perspective, researcher's cultural, and theoretical orientation. Overall, most of the JBI critical appraisal items could be assessed.

The included studies were all graded as low or moderate risk of bias. Please see Supplementary 2 for more details. We used the JBI Levels of Evidence to report the levels of evidence for the effectiveness of experimental designs (JBI, 2013). We obtained two Level 1.c evidence for two RCT (Da Silva et al., 2020; Oner et al., 2018), six Level 2.d evidence for non-RCTs with pre- and posttests (Doherty et al., 2016; Kent et al., 2015; Sawyer et al., 2013; Sayre et al., 2012a, 2012b; Shanks et al., 2020) and posttest studies (Omura et al., 2019a, 2019b), and one level 3 evidence for one qualitative study (Hanson et al., 2020).

3.4. Effects of assertive communication training on the attitudes, behaviors, and confidence levels of nurses for speaking up in cases of medical errors

We evaluated the effects of assertive communication training on the attitudes, behaviors, and confidence levels of nurses for speaking up in cases of medical errors. The levels of evidence (from high to low in two- and single-arm studies; Fig. 2) were also assessed.

3.4.1. Two-arm studies on assertive communication training to improve nurses' attitudes and behaviors for speaking up in cases of medical errors

The data of four studies (Da Silva et al., 2020; Omura et al., 2019b; Oner et al., 2018; Sayre et al., 2012a) with a two-arm study design (n = 380) was used to evaluate the training effects of assertive communication training on the attitudes and behaviors of nurses for speaking up in cases of medical errors (Da Silva et al., 2020; Omura et al., 2019b; Oner et al., 2018; Sayre et al., 2012a). In one of the studies nurses' attitudes towards speaking up were measured using a self-reported questionnaire (Sayre et al., 2012a). In the other three, speaking-up behavior was measured using structured tools (Da Silva et al., 2020; Omura et al., 2019b; Oner et al., 2018).

Regarding the effects of assertive communication training on speaking-up attitude score in the pooled results, the difference between the intervention and control groups at pretest was −0.41. After the training, the score related to speaking-up attitude was higher in the intervention group (SMD = 0.26; 95 % CI, −0.13 to 0.64) than in the control group. Regarding the effects of assertive communication training on speaking-up behavior in the pooled results, the intervention group exhibited a statistically significant higher score in speaking-up behavior than did the control group (SMD = 0.58; 95 % CI, 0.14–1.03). Moderate heterogeneity (I^2, 60 %) was detected in the pooled result (Fig. 2A).

3.4.2. Assertive communication training for improving nurses' attitudes towards speaking up in cases of medical errors

The data of 114 participants in two studies (Doherty et al., 2016; Sayre et al., 2012a) was used to evaluate the effects of speaking-up training on the attitudes of nurses at pre- and posttests (Doherty et al., 2016; Sayre et al., 2012a). Before assertive communication training, the speaking-up attitude score of nurses and nursing students was approximately 64.55 (95 % CI, 38.97–90.12) across the pooled results. After assertive communication training, the score was approximately 69.30 (95 % CI, 40.47–98.12). These findings were highly heterogeneous (I^2,

Table 1
Characteristics of the included studies.

Author (year)	Study design	Speaking up or confidence levels	Country, participants, and setting	Single group or experimental: control (number)	Interventions			Measurement		Results
					Teaching approaches	Duration	Content	Timing	Instruments	Significance
Sayre et al. (2012a, 2012b)	Quasiexperimental	Speak up	United States; registered nurses; two similar, acute care hospitals that belonged to the same health-care system	58:87	Discussion	5–6 h	Five real-life patient safety scenarios were discussed with nurses. The registered nurses could speak up when the situation impacted patient safety, followed by support from other nurses and physician leaders, discussion the organizational and individual obstacles that prevent the nurses from speaking up in cases of medical errors, and small group interactions after class where the nurses could implement their action plans for newly developed speaking-up behaviors.	Pre/post (1 or 2 months)	**Speaking-up assessment tool developed** by Premeaux and Bedeian (2003), **the Individual List of Nurse Behaviors Collaborative Practice Scale** developed by Weiss and Davis (1985)	Significant
Sawyer et al. (2013)	Quasiexperimental	Speak up	United States; Neonatal physicians, fellows, nurses, and respiratory therapists working in neonatal intensive care units	Single group (42)	Simulation and didactic	6 h	The courses were conducted as described in the TeamSTEPPS instructor guide, which included didactic training in teams and teamwork skills related to leadership, situation monitoring, mutual support (assertive statement; two-challenge rule; concerned, uncomfortable, and safety issue tool, and DESC), and communication. During the simulations, two medication order and performance errors were staged to assess the reactions of the health-care providers.	Pre/post	**Observers recorded** the nurses' willingness to speak up and challenge physicians in cases of medical errors (e.g., incorrect medication dose; not analyzed quantitatively)	Nonsignificant
Kent et al. (2015)	Quasiexperimental	Confidence levels	United States; senior-level nursing students	Single group (63)	Course (didactic and practical instructions)	N/F[a]	Using of standardized communication tools, such as "I am concerned...." or "I feel uncomfortable…," nurse–physician relationships, and strategies to address conflicts	Pre/post	**Health Professional Education in Patient Safety Survey** by Ginsburg et al. (2010)	Significant
Doherty et al. (2016)	Quasiexperimental	Speak up	United States; second-year nursing students	Single group (138)	Lecture, video presentations, group discussion, and role play	N/F[a]	TeamSTEPPS, voice concern: assertive statement; two-challenge rule; and concerned, uncomfortable, and safety issue tool	Pre/post (2 weeks)	**Nurse Workplace Scale** developed by DeMarco et al. (2008)	Significant
Oner et al. (2018)	RCT	Speak up	United States; labor and delivery and postpartum nurses	34 (L18:P16): 36 (L14:P22)	Discussion, role play, and simulation (standardized patient)	3 h	PowerPoint slides with major points, followed by role play, including assertive, submissive (passive), aggressive, and indirect (passive aggressive) behaviors; simulations; and debriefing. **Experimental:** assertiveness/	Pre/post	**The Rathus Assertiveness Schedule Scale** developed by Rathus (1973); **the Gerry Assertiveness Scale** modified by Begley and Glacken (2004); and the **Modified Pian–Smith Speaking -Up Evaluation Tool**	Nonsignificant

(*continued on next page*)

Table 1 (*continued*)

Author (year)	Study design	Speaking up or confidence levels	Country, participants, and setting	Single group or experimental: control (number)	Interventions			Measurement		Results
					Teaching approaches	Duration	Content	Timing	Instruments	Significance
							advocacy/concerned, uncomfortable, and safety issue tool/two-challenge rule with simulation. **Control:** illness severity, patient summary, action list, situational awareness, and contingency planning synthesis by receiver hand-off tool education with simulation.			
Omura et al. (2019a, 2019b)	Quasiexperimental	Speak up and confidence levels	Japan; third-year nursing students	73:75 (150)	Multimethod[b]	90 min	Nine-step process (Gagné et al., 2005) to address the conditions of learning and the cognitive domains indicated in Bloom's Revised Taxonomy by Anderson and Krathwohl (2001): speaking up in adverse events, crucial assertive communication to ensure patient safety, discussion, moral courage and patient advocacy, and rehearsal and practical assertive communication techniques, such as probe, alert, challenge and emergency action; conclusion, reason, example, and conclusion; and 5-step advocacy (attention getter, state your concern, state the problem, state a solution, and obtain agreement)	Post	**The Japanese version of the Rathus Assertive Schedule** tool developed by Suzuki et al. (2007); **the Theory of Planned Behavior–Assertive Communication Questionnaire** developed by the authors and constructed using the steps outlined by Francis et al. (2004); **and Satisfaction with Assertiveness Communication Training Program Survey**	Nonsignificant
Shanks et al. (2020)	Quasiexperimental	Confidence levels	United States; senior-level nursing students from a Midwestern University	31:33 (22:29)	Simulation	N/F[a]	A class on speaking up for patient safety and interprofessional communication based on the Agency for Healthcare Research and Quality TeamSTEPPS curriculum tools designed specifically situation, background, assessment, and recommendation as well as concerned, uncomfortable, and safety issue tools	Pre/post	**Student Health Professional Education in Patient Safety Survey**	Nonsignificant
Da Silva et al. (2020)	RCT	Speak up	Canadian; third-year nursing students	46:46 (44:39)	Virtual simulations	N/F[a]	Experimental: virtual simulations case study Control: paper case study	post	Two assessors observed the student utilized proportions of the two-challenge rule in a simulation scenario.	Significant
Hanson et al. (2020)	Qualitative questions and individual semi-structured interviews	Speak up	Australia; first-year nursing students	Single group (535)	Discussion and role play	N/F[a]	A video series of communication failures about unsafe practices that led to fatal medical complications, providing assertiveness frameworks: probe, alert, challenge, and emergency as well as concern, uncertain/	Post	**Seven qualitative survey questionnaires, individual semi-structured interviews, and thematic analysis**	NA

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Table 1 (*continued*)

Author (year)	Study design	Speaking up or confidence levels	Country, participants, and setting	Single group or experimental: control (number)	Interventions			Measurement		Results
					Teaching approaches	Duration	Content	Timing	Instruments	Significance
							uncomfortable, safety, and stop, a 10-min role play, and brief discussion			

[a] N/F: None found.
[b] Multimethod: prereading, a PowerPoint presentation, audiovisual recordings of authentic scenarios, didactic instructions, group discussion, and role play.

100 %). We expected a training score of >60, which indicates that the training improves nurses' attitudes for speaking up in cases of medical errors. No statistically significant differences were noted between pre- and posttests (p = .81; Fig. 2B).

3.4.3. Assertive communication training for improving nurses' times of speaking up in cases of medical errors

A total of 115 participants' data in three studies (Da Silva et al., 2020; Omura et al., 2019b; Sawyer et al., 2013) was used to evaluate the times of speaking up in cases of medical errors. Structured tools were used in these three studies to assess the times of speaking up in cases of medical errors. According to the pooled results, this rate was 38 % (95 % CI, 0.14–0.68) before the training, versus 78 % (95 % CI, 0.70–0.85) after the training, with very low heterogeneity (I^2, 0 %). We expected a training score of >60 %, which indicates that assertive communication training improves the speaking-up behaviors and attitudes of nurses. Statistically significant differences were noted in speaking-up performance between pre- and posttests (p < .01; Fig. 2C).

3.4.4. Assertive communication training for increasing nurses' confidence levels for speaking up in cases of medical errors

The data of three studies with 195 participants total was used to evaluate the effects of the assertive communication training on nursing students' confidence levels for speaking up in cases of medical errors (Kent et al., 2015; Omura et al., 2019a; Shanks et al., 2020). According to the pooled results, the mean score for their confidence level was 78.37 (95 % CI, 76.51–80.23) at pretest (Kent et al., 2015; Shanks et al., 2020), with low heterogeneity (I^2, 0 %), whereas it was 80.60 (95 % CI, 74.07–87.13) at posttest (Kent et al., 2015; Omura et al., 2019a; Shanks et al., 2020), with very high heterogeneity (I^2, 94 %). No statistically significant differences were noted in confidence levels between pre- and posttests (p = .52; Fig. 2D).

4. Discussion

In the present review, the research team revealed that assertive communication training resulted in statistically significantly improvement in the speaking-up behaviors of nurses and nursing students in cases of medical errors. No statistically significant improvement was noted in their attitude and confidence levels after training.

We identified a total of three potential key elements for effective assertive communication training: structured content, teaching approach, and sufficient intervention time. In most studies the content of assertive communication training was developed using techniques related to structured communication (Doherty et al., 2016; Hanson et al., 2020; Kent et al., 2015; Omura et al., 2019a; Oner et al., 2018; Sawyer et al., 2013; Shanks et al., 2020). In a systematic review, authors reported that teamwork skills and structured communication techniques were used in assertive communication training in some studies (Omura et al., 2017). The educators used teamwork skills and structured communication techniques, such as the use of an assertive statement; two-challenge rule; CUS (concerned, uncomfortable, and safety issue) as well as DESC (describe, express, suggest, and consequences) from the Agency for Healthcare Research and Quality TeamSTEPPS (Doherty et al., 2016; Kent et al., 2015); and PACE (probe, alert, challenge, and emergency framework) (Hanson et al., 2020), could help devise adaptable strategies for learners to practice and incorporate into their behaviors. This may improve the learners' speaking-up performance in cases of medical errors.

Most studies used multiple teaching approaches. Role play and simulation were identified as being effective in improving the leaners' speaking-up performance (Doherty et al., 2016; Oner et al., 2018; Sawyer et al., 2013; Shanks et al., 2020). Sayre et al. (2012a) adopted a strategy involving medical error–related scenario discussion to encourage the learners to share their personal reflections and to facilitate peer support. Such active learning interventions may lead to

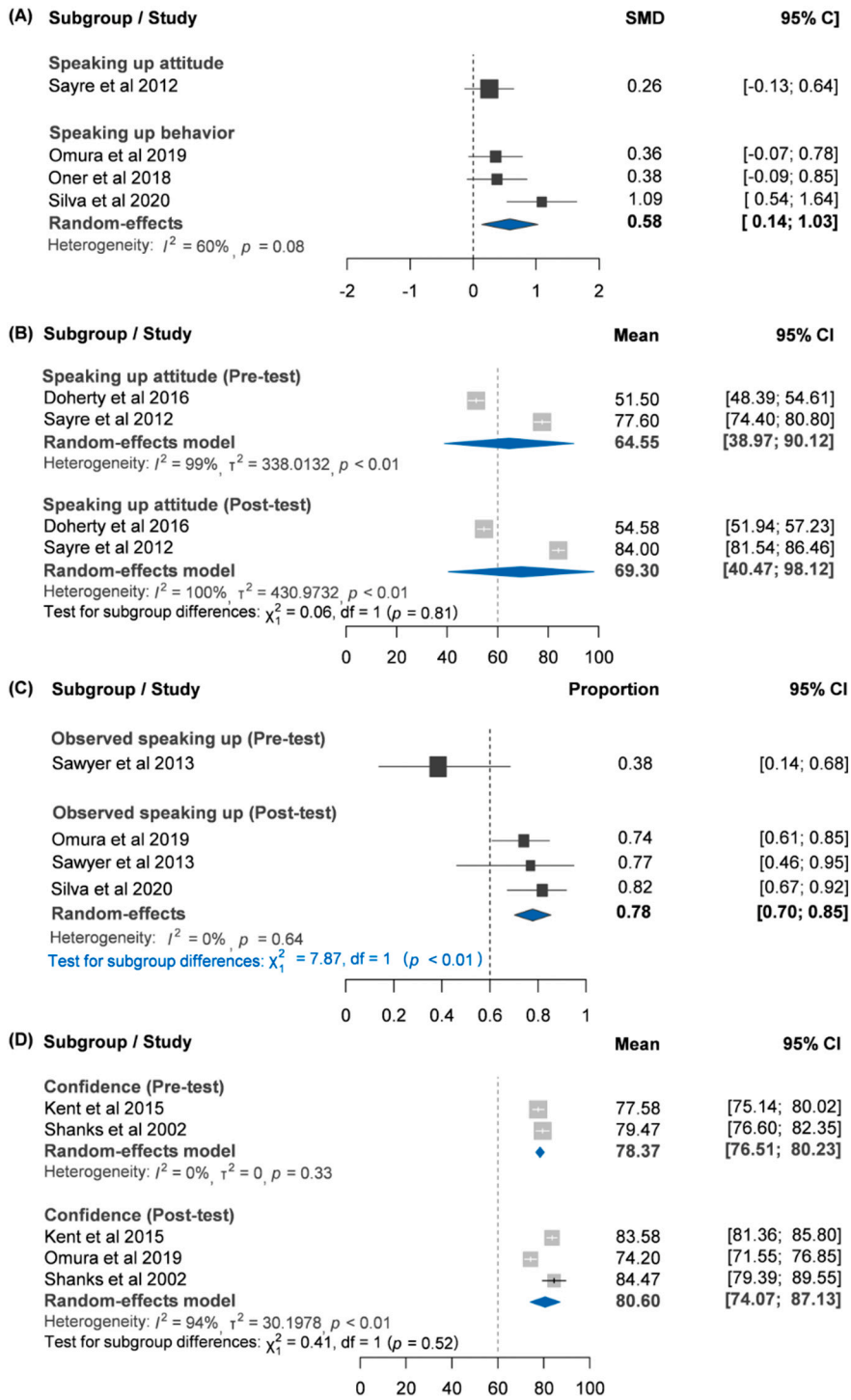

Fig. 2. Forest plots for two-arm studies on assertive communication training to improve the (A) attitudes and behaviors, (B) attitudes, (C) times of speaking-up, (D) confidence levels of nurses for speaking up in cases of medical errors. CI, confidence interval; SMD, standardized mean difference.

positive outcomes (Prince, 2004). Additionally, the use of multiple teaching approaches is a good strategy for ensuring continual learner attention (Bunce et al., 2010) and obtaining positive learning outcomes. When developing assertive communication training for nursing professionals to encourage speaking up in the event of medical error, educators should consider implementing multiple teaching strategies with active learning models, such as interactive discussion or hands-on practice activities. To develop more effective and efficient assertive communication training, future studies are needed to investigate the most effective and efficient combination of teaching approaches.

Although there was no empirical evidence to indicate the minimum duration of time for an effective assertive communication training program, the influence of intervention duration should be considered when designing the education program. The intervention duration of the 11 studies ranged from 90 min to 6 h, but studies with statistically insignificant results all had intervention duration <3 h (Omura et al., 2019a, 2019b; Oner et al., 2018). In assertive communication training, an adequate intervention time is essential for the trainees to learn and practice various nontechnical skills, which may lead to positive outcomes. The speaking-up behavior of nurses in cases of medical errors is a

complex response. It requires not only adequate knowledge but also confidence and nontechnical skills. Sufficient training time is required for repetitive practice and learning process adjustment for the trainees to reach adequate performance levels and attain effective behavior changes. Thus, adequate training time may lead to improved performance in the workplace (Lacerenza et al., 2017) and improved behaviors and skills (Taylor et al., 2005). The educators should consider the various characteristics of the learners, such as background knowledge, experience, or the baseline ability of non-technical skills to estimate adequate intervention time for positive learning outcomes.

The Kirkpatrick Model was used to evaluate the outcomes of educational programs. In healthcare education, the higher Kirkpatrick's level indicates that the learner's actual behavior changes would influence patients' or even the organizations' outcomes. One of the eleven study outcomes was related to the improvement of students' satisfaction (Level 1 learning reaction/satisfaction) (Omura et al., 2019a). The rest of the studies examined the learners' attitudes (Doherty et al., 2016; Sayre et al., 2012a, 2012b; Hanson et al., 2020), confidence levels (Kent et al., 2015; Shanks et al., 2020), and behavioral changes in scenarios (Da Silva et al., 2020; Sawyer et al., 2013; Oner et al., 2018; Omura et al., 2019b). All of those studies had level 2 learning outcomes. Currently, no study investigates the changes of learners' actual behaviors in the real clinical environment (Level 3) and the influences of the learners' behaviors on the targeted patient safety outcomes (Level 4), indicating areas for future study. The purpose of assertive communication training is to help learners to perform speaking up behaviors when faced with medical error events in their daily practice, and then prevent or minimize patient harm. Nurses' actual behavior changes and patient safety outcomes should be evaluated in order to guide the future direction and educational content assertive communication training programs.

5. Strengths and limitations

The strengths of the present meta-analysis are as follows. Rigorous assessment of the methodical quality of the included studies revealed high internal validity. We included not only two-arm but also single-arm studies, which increased the external validity of the review. Furthermore, the analysis process was standardized, and the findings were interpreted with caution.

Our study has some limitations. Some of the included studies had two-arm and RCT design; more RCTs should be conducted in the future. Second, the included studies recruited nursing professionals at various levels of their nursing career (nurses and nursing students). The effects of assertive communication training may vary widely across these nursing professionals; subgroup analysis was not possible because of the low number of the included studies. Finally, the effectiveness of different intervention duration times might be tested in future studies to identify the most effective length.

6. Conclusions

Assertive communication training appears to be an effective strategy for improving the speaking-up behavior of nurses in cases of medical errors. Effective and comprehensive assertive communication training must incorporate structured steps to personalize, and the adequate training duration and to communicate relevant feedback to nurses through real-world case discussion or simulation practice.

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CRediT authorship contribution statement

Chen HW, Chiu YJ, and Kang YN were involved in study selection, quality assessment, data extraction, collection; Chen HW and Kang YN were involved in study software, formal analysis, visualization; Chen HW, Wu JC, and Kang YN were completed the visualization writing - original draft, writing - review & editing of the manuscript; Hu SH and Wu JC were involved in conceptualization, quality assessment and reviewing of the manuscript.

Declaration of competing interest

There are no conflicts of interest to declare.

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References

AHRQ Agency for Healthcare Research and Quality, 2019. TeamSTEPPS 2.0 Fundamentals, Rockville, MD. Available at: https://www.ahrq.gov/teamstepps/instructor/fundamentals/index.html. Accessed 4th Sep 2021.

Anderson, L.W., Krathwohl, D.R., 2001. A taxonomy for learning, teaching and assessing: A revision of Bloom's taxonomy of educational objectives. Longman, New York, NY.

Austin, M., Derk, J., 2016. Lives lost, lives saved: A comparative analysis of avoidable deaths at hospitals graded by The Leapfrog Group. Armstrong Institute for Patient Safety and Quality. Available at: https://www.hospitalsafetygrade.org/media/file/LivesLostLivesSavedWhitePaper.pdf. Accessed 4th Sep 2021.

Begley, C.M., Glacken, M., 2004. Irish nursing students' changing levels of assertiveness during their pre-registration programme. Nurse Educ. Today 24 (7), 501–510. https://doi.org/10.1016/j.nedt.2004.06.002.

Best, J.A., Kim, S., 2019. The FIRST curriculum: cultivating speaking up behaviors in the clinical learning environment. J. Contin. Educ. Nurs. 50 (8), 355–361. https://doi.org/10.3928/00220124-20190717-06.

Bunce, D.M., Flens, E.A., Neiles, K.Y., 2010. How long can students pay attention in class? A study of student attention decline using clickers. J. Chem. Educ. 87 (12), 1438–1443. https://doi.org/10.1021/ed100409p.

Da Silva, C., Peisachovich, E., Anyinam, C.K., Coffey, S., Graham, L., Tavangar, F., 2020. Speaking up against hierarchy: a simulation geared towards nursing students. Cureus 12 (12), e11977. https://doi.org/10.7759/cureus.11977.

DeMarco, R., Roberts, S.J., Norris, A., McCurry, M.K., 2008. The development of the Nurse Workplace Scale: self-advocating behaviors and beliefs in the professional workplace. J. Prof. Nurs. 24 (5), 296–301. https://doi.org/10.1016/j.profnurs.2007.10.009.

Doherty, C., Landry, H., Pate, B., Reid, H., 2016. Impact of communication competency training on nursing students' self-advocacy skills. Nurse Educ. 41 (5), 252–255. https://doi.org/10.1097/nne.0000000000000274.

Francis, J., Eccles, M.P., Johnston, M., Walker, A.E., Grimshaw, J.M., Foy, R., Kaner, E.F.S., Smith, L., Bonetti, D., 2004. Constructing questionnaires based on the theory of planned behaviour: A manual for health services researchers. Centre for Health Services Research, University of Newcastle upon Tyne, Newcastle upon Tyne, UK.

Gagné, R.M., Wager, W.W., Golas, K.C., Keller, J.M., 2005. Principles of instructional design, 5th. Wadsworth/Thomson Learning, Belmont, CA.

Ginsburg, L., Castel, E., Tregunno, D., Norton, P., Medves, J., Campbell, H., 2010. H-PEPSS: Health Professional Education in Patient Safety Survey. http://www.yorku.ca/patientsafety/H-PEPSS/researchers.htm.

Green, B., Oeppen, R.S., Smith, D.W., Brennan, P.A., 2017. Challenging hierarchy in healthcare teams - ways to flatten gradients to improve teamwork and patient care. Br. J. Oral Maxillofac. Surg. 55 (5), 449–453. https://doi.org/10.1016/j.bjoms.2017.02.010.

Hanson, J., Walsh, S., Mason, M., Wadsworth, D., Framp, A., Watson, K., 2020. 'Speaking up for safety': a graded assertiveness intervention for first year nursing students in preparation for clinical placement: thematic analysis. Nurse Educ. Today 84, 104252. https://doi.org/10.1016/j.nedt.2019.104252.

Higgins, J.P.T., Thomas, J., Chandler, J., Cumpston, M., Li, T., Page, M.J., Welch, V.A. (Eds.), 2022. Cochrane Handbook for Systematic Reviews of Interventions (Updated February 2022), 6.3. www.training.cochrane.org/handbook. (Accessed 28 October 2021).

Kent, L., Anderson, G., Ciocca, R., Shanks, L., Enlow, M., 2015. Effects of a senior practicum course on nursing students' confidence in speaking up for patient safety. J. Nurs. Educ. 54 (3 Suppl), S12–S15. https://doi.org/10.3928/01484834-20150218-04.

Kirkpatrick, D.L., 2009. Evaluating training programs: the four levels. ReadHowYouWant.com. Available at: https://books.google.com.tw/books?id=4rYs1wnyUHUC. Accessed 6th Oct 2021.

Lacerenza, C.N., Reyes, D.L., Marlow, S.L., Joseph, D.L., Salas, E., 2017. Leadership training design, delivery, and implementation: a meta-analysis. J. Appl. Psychol. 102 (12), 1686.

Makary, M.A., Daniel, M., 2016. Medical error-the third leading cause of death in the US. BMJ 353, i2139. https://doi.org/10.1136/bmj.i2139.

Maxfield, D., Grenny, J., Lavandero, R., Groah, L., 2011. The Silent Treatment: Why Safety Tools and Checklists Aren't Enough. Patient Safety & Quality Healthcare. Available at: https://faculty.medicine.umich.edu/sites/default/files/resources/silent_treatment.pdf. Accessed 10th Sep 2021.

Moher, D., Liberati, A., Tetzlaff, J., Altman, D.G., Group, P., 2009. Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. PLoS Medicine 6 (7), e1000097.

Okuyama, A., Wagner, C., Bijnen, B., 2014. Speaking up for patient safety by hospital-based health care professionals: a literature review. BMC Health Serv. Res. 14 (1), 61. https://doi.org/10.1186/1472-6963-14-61.

Omura, M., Maguire, J., Levett-Jones, T., Stone, T.E., 2017. The effectiveness of assertiveness communication training programs for healthcare professionals and students: a systematic review. Int. J. Nurs. Stud. 76, 120–128. https://doi.org/10.1016/j.ijnurstu.2017.09.001.

Omura, M., Levett-Jones, T., Stone, T.E., 2019a. Design and evaluation of an assertiveness communication training programme for nursing students. J. Clin. Nurs. 28 (9–10), 1990–1998. https://doi.org/10.1111/jocn.14813.

Omura, M., Levett-Jones, T., Stone, T.E., 2019b. Evaluating the impact of an assertiveness communication training programme for Japanese nursing students: a quasi-experimental study. Nurs. Open 6 (2), 463–472. https://doi.org/10.1002/nop2.228.

Oner, C., Fisher, N., Atallah, F., Son, M.A., Homel, P., Mykhalchenko, K., Minkoff, H., 2018. Simulation-based education to train learners to "speak up" in the clinical environment: results of a randomized trial. Simul. Healthc. 13 (6), 404–412. https://doi.org/10.1097/sih.0000000000000335.

Premeaux, S.F., Bedeian, A.G., 2003. Breaking the silence: The moderating effects of self-monitoring in predicting speaking up in the workplace. J. Manag. Stud. 40 (6), 1537–1562. https://doi.org/10.1111/1467-6486.00390.

Prince, M., 2004. Does active learning work? A review of the research. J. Eng. Educ. 93 (3), 223–231. https://doi.org/10.1002/j.2168-9830.2004.tb00809.x.

Raica, D.A., 2009. Effect of action-oriented communication training on nurses' communication self-efficacy. Medsurg Nurs. 18 (6), 343–360.

Rathus, S.A., 1973. A 30-item schedule for assessing assertive behavior. Behav. Ther. 4 (3), 398–406. https://doi.org/10.1016/S0005-7894(73)80120-0.

Reason, J., 2005. Safety in the operating theatre - part 2: human error and organisational failure. Qual. Saf. Health Care 14, 56–60. https://doi.org/10.1016/S0953-7112(05)80010-9.

Sawyer, T., Laubach, V.A., Hudak, J., Yamamura, K., Pocrnich, A., 2013. Improvements in teamwork during neonatal resuscitation after interprofessional TeamSTEPPS training. Neonat. Netw. 32 (1), 26–33. https://doi.org/10.1891/0730-0832.32.1.26.

Sayre, M.M., McNeese-Smith, D., Leach, L.S., Phillips, L.R., 2012a. An educational intervention to increase "speaking-up" behaviors in nurses and improve patient safety. J. Nurs. Care Qual. 27 (2), 154–160. https://doi.org/10.1097/NCQ.0b013e318241d9ff.

Sayre, M.M., McNeese-Smith, D., Phillips, L.R., Leach, L.S., 2012b. A strategy to improve nurses speaking up and collaborating for patient safety. J. Nurs. Adm. 42 (10), 458–460. https://doi.org/10.1097/NNA.0b013e31826a1e8a.

Schwappach, D.L.B., Gehring, K., 2014. 'Saying it without words': a qualitative study of oncology staff's experiences with speaking up about safety concerns. BMJ Open 4 (5), e004740. https://doi.org/10.1136/bmjopen-2013-004740.

Schwappach, D., Richard, A., 2018. Speak up-related climate and its association with healthcare workers' speaking up and withholding voice behaviours: a cross-sectional survey in Switzerland. BMJ Qual. Saf. 27 (10), 827–835. https://doi.org/10.1136/bmjqs-2017-007388.

Shanks, L.C., Chiu, S., Zelko, M., Fleming, E., Germano, S., 2020. Speaking up to authority in a simulated medication error scenario. Clin. Simul. Nurs. 45, 28–31. https://doi.org/10.1016/J.Ecns.2020.01.008.

Suzuki, E., Kanoya, Y., Katsuki, T., Sato, C., 2007. Verification of reli- ability and validity of a Japanese version of the Rathus Assertiveness Schedule. J. Nurs. Manag. 15 (5), 530–537. https://doi.org/10.1111/j.1365-2834.2007.00691.x.

Taylor, P.J., Russ-Eft, D.F., Chan, D.W., 2005. A meta-analytic review of behavior modeling training. J. Appl. Psychol. 90 (4), 692–709. https://doi.org/10.1037/0021-9010.90.4.692.

The Joanna Briggs Institute, 2013. JBI Levels of Evidence. Available at: https://jbi.global/sites/default/files/2019-05/JBI-Levels-of-evidence_2014_0.pdf. Accessed 4th Sep 2021.

Viechtbauer, W., 2010. Conducting meta-analyses in R with the metafor package. J. Stat. Softw. 36 (3), 1–48. https://doi.org/10.18637/jss.v036.i03.

Weiss, S.J., Davis, H.P., 1985. Validity and reliability of the Collaborative Practice Scales. Nurs. Res. 34 (5), 299–305. https://doi.org/10.1097/00006199-198509000-00010.